Pricing and hedging financial derivatives: Risk-neutral valuation

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Bird's eye view and key concepts of this lecture

- *Pricing and hedging financial derivatives in a specific probabilistic model of the financial market*
- *Classical setting: Brownian filtration, Itô processes*
- *Standard asumptions: perfect markets, everything can be hedged*
- *Dynamics of a self-financing portfolio containing several risky assets. Representation of portfolio strategies in terms of the number of units of each asset, the amount invested in each asset, and the proportion of total wealth invested in each asset.*
- *Admissible strategies, absence of arbitrage opportunities, risk premium.*
- *Risk neutral valuation formula for contingent claims.*

Key assumptions and modeling choices

- **There are no riskless arbitrage opportunities**
  (impossible to outperform risk-free asset or earn money from zero initial investment without taking any risk)
  $\Rightarrow$ True in normal market conditions
  (but remember transaction costs)

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- **No transactions costs or taxes**

  $\Rightarrow$ False

  $\Rightarrow$ There are multiple transaction costs: bid-ask spreads, exchange fees, order book liquidity costs
  For liquid assets and low hedging frequencies costs are small

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- **The short selling of securities is permitted**
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- **There exists a risk-free asset**
  $\Rightarrow$ Some government bonds and short-term collateralized interbank loans may be considered risk-free $\Rightarrow$ True
  (Remember subprime crisis and Eurozone crisis)

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  $\Rightarrow$ False
- **The market is complete: all risk may be hedging by investing in risky assets**
  $\Rightarrow$ In practice there are unhedged risks (stochastic volatility)

  $\Rightarrow$ False

The model for asset prices

Let $W = (W_t^1, \dots, W_t^d)_{t\geq 0}^{\mathrm{T}}$ be a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$, and $(\mathcal{F}_t)_{t\geq 0}$ its completed natural filtration.

Consider a financial market with a risk-free asset

$$S_t^0 = \exp\left(\int_0^t r(u)du\right),$$

and d risky assets

$$S_t^i = S_0^i \exp\left(\int_0^t \left(b^i(u) - \frac{1}{2}\sum_{j=1}^d |\sigma^{ij}(t)|^2\right) dt + \int_0^t \sum_{j=1}^d \sigma^{ij}(t) dW_t^j\right), \quad i = 1 \dots d,$$

where r, μ and σ are adapted process satisfying

$$\int_0^T (\|r(t)\| + \|b(t)\| + \|\sigma(t)\|^2) dt < \infty, \quad \text{p.s.},$$

We assume that $\sigma(t)$ is invertible and $r(t) \geq 0$ for all t.

The model for asset prices

By Itô's formula,

$$dS_t^i = S_t^i \left(b^i(t)dt + \sum_{i=1}^{d} \sigma^{ij}(t)dW_t^j \right),$$

or with matrix notation

$$dS_t = \mathrm{diag}[S_t](b(t)dt + \sigma(t)dW_t),$$

where $S_t = (S_t^1, \dots, S_t^d)^{\mathrm{T}}$, $b(t) = (b^1(t), \dots, b^d(t))^{\mathrm{T}}$.

Introduce the *risk premium vector*: $\lambda(t) = \sigma(t)^{-1}(b(t) - r(t)\mathbf{1})$, so that

$$dS_t = \mathrm{diag}[S_t] \left\{ r(t)\mathbf{1}dt + \sigma(t)(\lambda(t)dt + dW_t) \right\}.$$

To simplify notation, use discounting

$$\tilde{S}_t = \frac{S_t}{S_t^0} = S_t e^{-\int_0^t r(s)ds} \quad \Rightarrow \quad d\tilde{S}_t = \mathrm{diag}[\tilde{S}_t]\sigma(t)(\lambda(t)dt + dW_t).$$

Portfolio strategies

A strategy can be expressed

- in terms of *quantities* of each risky asset $\delta_t = (\delta_t^1, \dots, \delta_t^d) \in \mathbb{R}^d$,
- in terms of *amounts* to invest in each risky asset $\pi_t = (\pi_t^1, \dots, \pi_t^d) \in \mathbb{R}^d$
- if the portfolio value X_t is positive at all times, in terms of *proportions* of wealth invested in each risky asset $\omega_t = (\omega_t^1, \dots, \omega_t^d) \in \mathbb{R}^d$.

These coefficient are related as follows:

$$\pi_t^i = \delta_t^i S_t^i \quad \text{et} \quad \omega_t^i = \frac{\delta_t^i S_t^i}{X_t}, \quad i = 1, \dots, d.$$

The amount invested into risk-free asset is given by,

$$X_t - \sum_{i=1}^d \delta_t^i S_t^i = X_t - \sum_{i=1}^d \pi_t^i = X_t \left(1 - \sum_{i=1}^d \omega_t^i\right)$$

Self-financing strategies

The self-financing equation expresses the fact that no amount is withdrawn or injected into the portfolio.

- With quantities:

$$dX_t = \sum_{i=1}^{d} \delta_t^i dS_t^i + \left(X_t - \sum_{i=1}^{d} \delta_t^i S_t^i \right) \frac{dS_t^0}{S_t^0}.$$

- With amounts:

$$dX_t = \sum_{i=1}^{d} \pi_t^i \frac{dS_t^i}{S_t^i} + \left(X_t - \sum_{i=1}^{d} \pi_t^i \right) \frac{dS_t^0}{S_t^0}.$$

- With proportions:

$$\frac{dX_t}{X_t} = \sum_{i=1}^{d} \omega_t^i \frac{dS_t^i}{S_t^i} + \left(1 - \sum_{i=1}^{d} \omega_t^i \right) \frac{dS_t^0}{S_t^0}.$$

Portfolio strategies

From now on we focus on the amounts.

A strategy is a pair $(x, (\pi_t)_{0\leq t\leq T})$, where x is the initial value and $\pi_t = (\pi_t^1, \dots, \pi_t^d) \in \mathbb{R}^d$ are the amounts to invest in risky assets.

With discounting::

$$d\tilde{X}_t = \tilde{\pi}_t \mathrm{diag}[\tilde{S}_t]^{-1} d\tilde{S}_t = \tilde{\pi}_t \sigma(t)\{\lambda(t)dt + dW_t\}.$$

We denote by $X_t^{x,\pi}$ the value of portfolio with initial value x and strategy π.

Risk-neutral probability

Assume Novikov's condition:

$$\mathbb{E}\left[\exp\left(\frac{1}{2}\int_0^T \|\lambda(t)\|^2 dt\right)\right] < \infty,$$

then under the measure $\mathbb{Q}$ defined by

$$\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = \exp\left(-\int_0^T \lambda(t) dW_t - \frac{1}{2}\int_0^T \|\lambda(t)\|^2 dt\right),$$

the process $\widehat{W}_t = \int_0^t \lambda(s)ds + W_t$ is a Brownian motion and we write

$$\tilde{X}_t^{x,\pi} = x + \int_0^t \tilde{\pi}_u \sigma(u) d\widehat{W}_u.$$

Under $\mathbb{Q}$, the discounted values of assets and portfolios are local martingales, but not always true martingales.

Admissible strategies

Definition

The strategy $(\pi_t)_{0\leq t\leq T}$ is said to be admissible if

i. the stochastic integral is well defined:

$$\int_0^T \|\tilde{\pi}_u \sigma(u)\|^2 du < \infty \quad \text{p.s.,}$$

ii. the portfolio value satisfies

$$\tilde{X}_t^{0,\pi} \geq M_t, \quad 0 \leq t \leq T,$$

where M is an a.s. continuous $\mathbb{Q}$-martingale (finite credit line condition).

The set of admissible strategies is denoted $\mathcal{A}$.

Doubling strategies

The “finite credit line condition” is needed to avoid “doubling strategies”.

Assume that the interest rate is null and risky asset follows $S_t = W_t$.

We hold $2^{n/2}$ units of risky asset on the interval $(t_{k-1}, t_k]$, where $t_k = 1 - \frac{1}{2^k}$.

The portfolio value at t_n is

$$X_{t_n} = \sum_{k=1}^{n} 2^{k/2}(W_{t_k} - W_{t_{k-1}}) = \sum_{k=1}^{n} Z_k,$$

where $(Z_k)_{k\geq 1}$ is a sequence of i.i.d. $N(0,1)$.

$(X_{t_n})_{n\geq 1}$ is a Gaussian random walk which satisfies

$$\inf\{n : X_{t_n} \geq K\} < \infty \quad \text{p.s.} \quad \forall K.$$

This strategy is not admissible.

Absence of arbitrage

Definition (Absence d'opportunité d'arbitrage)

The market does not admit arbitrage opportunity if for any admissible strategy π,

$$X_T^{0,\pi} \geq 0 \quad \text{p.s.} \quad \Rightarrow \quad X_T^{0,\pi} = 0 \quad \text{p.s.}$$

Theorem

The financial market defined above does not admit arbitrage opportunity

Absence of arbitrage: proof

Let π be an admissible strategy with $X^{0,\pi}_T \geq 0$ a.s., and define

$$\tau_n = \inf\left\{t : \int_0^t \|\tilde\pi_u \sigma(u)\|^2 du \geq n\right\} \wedge T.$$

Then, (τ_n) converges a.s. to T and

$$\mathbb{E}^{\mathbb{Q}}\left[\int_0^{\tau_n} \|\tilde\pi_u \sigma(u)\|^2 du\right] \leq n < \infty,$$

so that

$$\mathbb{E}^{\mathbb{Q}}[\tilde X^{0,\pi}_{\tau_n}] = 0.$$

for all n. By Fatou's lemma, since $\tilde X^{0,\pi}_{\tau_n} - M_{\tau_n} \geq 0$, we have,

$$\mathbb{E}^{\mathbb{Q}}[\tilde X^{0,\pi}_T] = \mathbb{E}^{\mathbb{Q}}[\lim_{n\to\infty}(\tilde X^{0,\pi}_{\tau_n} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] \leq \lim_{n\to\infty} \mathbb{E}^{\mathbb{Q}}[(\tilde X^{0,\pi}_{\tau_n} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] = 0,$$

so that $X^{0,\pi}_T = 0$ a.s.

Superhedging price

Superhedging price: minimum cost of a dominating portfolio.

Definition (Superhedging / subhedging price)

Let $G \in \mathcal{F}_T$. The superhedging price of G at $t = 0$ and $t \in [0, T]$ is defined by

$$\overline{V}_0(G) = \inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \geq G\},$$

$$\overline{V}_t(G) = \operatorname{essinf}\{X \in \mathcal{F}_t : \exists \pi \in \mathcal{A} \text{ with } X_t^{x,\pi} = X \text{ and } X_T^{x,\pi} \geq G\}.$$

Similarly, the subhedging price is defined by

$$\underline{V}_0(G) = \sup\{x \in \mathbb{R} : \exists \pi \text{ with } -\pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \leq G\},$$

$$\underline{V}_t(G) = \operatorname{esssup}\{X \in \mathcal{F}_t : \exists \pi \text{ with } -\pi \in \mathcal{A}, X_t^{x,\pi} = X \text{ and } X_T^{x,\pi} \leq G\}.$$

(attention to the minus sign in the subhedging price)

Properties of the superhedging price

1. Monotonicity: if $G_1 \geq G_2$ a.s., then $\overline{V}(G_1) \geq \overline{V}(G_2)$.
2. Sub-additivity: for all G_1, G_2, $\overline{V}(G_1 + G_2) \leq \overline{V}(G_1) + \overline{V}(G_2)$.
3. Relationship with the subhedging price: for all G, $\underline{V}(G) = -\overline{V}(-G)$ and $\underline{V}(G) \leq \overline{V}(G)$.
4. In the absence of arbitrage, the market price of a pay-off G, denoted by $V(G)$, satisfies $\underline{V}(G) \leq V(G) \leq \overline{V}(G)$.

$\Rightarrow$ If a claim G is replicable with a portfolio (x, π) such that both π and $-\pi$ are admissible, then $V(G) = x$.

Proof: monotonicity

From the definition of the superhedging price, for any $\varepsilon > 0$, one can find $x \in \mathbb{R}$ and $\pi \in \mathcal{A}$ such that $x \leq \overline{V}(G_1) + \varepsilon$ and

$$X_T^{x,\pi} \geq G_1.$$

Thus also, $X_T^{x,\pi} \geq G_2$, so that $\overline{V}(G_2) \leq x \leq \overline{V}(G_1) + \varepsilon$. Since ε is arbitrary, we conclude that $\overline{V}(G_2) \leq \overline{V}(G_1)$.

Proof: sub-additivity

From the definition of the superhedging price, for any $\varepsilon > 0$ we can find $x_1, x_2 \in \mathbb{R}$ and $\pi_1, \pi_2 \in \mathcal{A}$ such that

$$x_1 \leq \overline{V}(G_1) + \varepsilon, \quad x_2 \leq \overline{V}(G_2) + \varepsilon, \quad X_T^{x_1,\pi_1} \geq G_1 \text{ and } X_T^{x_2,\pi_2} \geq G_2.$$

By linearity of portfolio,

$$X_T^{x_1+x_2,\pi_1+\pi_2} \geq G_1 + G_2,$$

so that

$$\overline{V}(G_1 + G_2) \leq x_1 + x_2 \leq V(G_1) + V(G_2) + 2\varepsilon.$$

Since ε is arbitrary, we conclude that

$$\overline{V}(G_1 + G_2) \leq x_1 + x_2 \leq V(G_1) + V(G_2).$$

Proof: relationship with the superhedging price

By definition of the subhedging price,

$$\begin{aligned}\underline{V}(G) &= \sup\{x \in \mathbb{R} : \exists \pi \text{ with } -\pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \leq G\}\\ &= -\inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{-x,-\pi} \leq G\}\\ &= -\inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \geq -G\} = -\overline{V}(-G).\end{aligned}$$

By sub-additivity,

$$\overline{V}(G) - \underline{V}(G) = \overline{V}(G) + \overline{V}(-G) \geq \overline{V}(G - G) = \overline{V}(0) = 0.$$

where the last equality is due to AOA

Pricing replicable claims

- If a contingent claim can be both bought and sold at price $V(G)$, clearly arbitrage can be constructed if $V(G) \notin [\underline{V}(G), \overline{V}(G)]$.
- If a claim G is replicable with a portfolio (x, π),

$$\overline{V}(G) \leq x.$$

- If $-\pi$ is admissible, then also $\underline{V}(G) \geq x$.
- Since $\overline{V}(G) \geq \underline{V}(G)$, we conclude that $\underline{V}(G) = \overline{V}(G) = V(G) = x$.

Pricing contingent claims

Theorem

Let $G \in \mathcal{F}_T$ be a contingent claim with $\widetilde{G} \in L^2(\mathcal{F}_T, \mathbb{Q})$. Then there exists a replicating portfolio for G and its no arbitrage price is given by

$$V_t(G) = \mathbb{E}^Q\left[e^{-\int_t^T r(s)ds} G \middle| \mathcal{F}_t\right].$$

By martingale representation, there is a unique adapted process $H \in \mathcal{H}^2$ with

$$\widetilde{G} = \mathbb{E}[\widetilde{G}] + \int_0^T H_s d\widehat{W}_s.$$

Let $\tilde{\pi}_t = \sigma(t)^{-1} H_t$. Then

$$\widetilde{X}_t^{x,\pi} := \mathbb{E}[\widetilde{G}] + \int_0^t \tilde{\pi}_s \sigma(s) d\widehat{W}_s$$

is the discounted value of a self-financing portfolio which replicates $\widetilde{G}$.
Moreover $\widetilde{X}_t$ is a $\mathbb{Q}$-martingale hence it is admissible.

Summary

Pricing contingent claims

In the model of this lecture, the unique no-arbitrage price of an option with pay-off $G \in L^2(\mathcal{F}_T, \mathbb{Q})$ at time t is given by

$$V_t(G) = \mathbb{E}^Q \left[e^{-\int_t^T r(s)ds} G \middle| \mathcal{F}_t \right],$$

where $\mathbb{Q}$ is the risk-neutral probability, under which the risky assets satisfy

$$dS_t = \mathrm{diag}[S_t](r(t)\mathbf{1}dt + \sigma(t)d\widehat{W}_t),$$

where $\widehat{W}$ is a $\mathbb{Q}$-brownian motion.

Black-Scholes formula

Assume that there is a single risky asset, the pay-off is $G = g(S_T)$, and r and σ are deterministic. The pricing formula becomes

$$V_t(G) = \mathbb{E}^Q\left[e^{-\int_t^T r(s)ds} g(S_T)\Big|\mathcal{F}_t\right]$$
$$= \mathbb{E}^Q\left[e^{-\int_t^T r(s)ds} g\left(S_t e^{\int_t^T \left(r(s)-\frac{\sigma^2(s)}{2}\right)ds + \int_t^T \sigma(s)d\widehat{W}_s}\right)\Big|\mathcal{F}_t\right] := v(t, S_t),$$

or

$$v(t,S) = \mathbb{E}^Q\left[e^{-\int_t^T r(s)ds} g\left(S e^{\int_t^T \left(r(s)-\frac{\sigma^2(s)}{2}\right)ds + \int_t^T \sigma(s)d\widehat{W}_s}\right)\right]$$
$$= \mathbb{E}^Q\left[e^{-(T-t)\bar{r}} g\left(S e^{\left(\bar{r}-\frac{\bar{\sigma}^2}{2}\right)(T-t)+\bar{\sigma}\widehat{W}_{T-t}}\right)\right],$$

where

$$\bar{r} := \frac{1}{T-t}\int_t^T r(s)ds \quad \text{et} \quad \bar{\sigma}^2 = \frac{1}{T-t}\int_t^T \sigma^2(s)ds.$$

Black-Scholes formula: call option

For the call option $g(S) = (S - K)^+$ and

$$V_t(G) = \mathbb{E}^Q\left[e^{-\int_t^T r(s)ds}(S_T - K)^+ \middle| \mathcal{F}_t\right]$$
$$= e^{\int_0^t r(s)ds}\mathbb{E}^Q\left[Z_T \mathbf{1}_{S_T \geq K} \middle| \mathcal{F}_t\right] - Ke^{-\int_t^T r(s)ds}\mathbb{Q}[S_T \geq K|\mathcal{F}_t]$$

with

$$Z_T = e^{-\int_0^T \frac{\sigma^2(s)}{2}ds + \int_0^T \sigma(s)d\widehat{W}_s}.$$

Introduce a new probability (share measure) $\widetilde{\mathbb{Q}}$ given by

$$\frac{d\widetilde{\mathbb{Q}}}{d\mathbb{Q}}\Big|_{\mathcal{F}_T} = Z_T, \quad \text{such that} \quad \widetilde{W}_t = \widehat{W}_t - \int_0^t \sigma(s)ds$$

is a $\tilde{Q}$-Brownian motion. Then,

$$V_t(G) = S_t\widetilde{\mathbb{Q}}\left[S_T \geq K|\mathcal{F}_t\right] - Ke^{-\int_t^T r(s)ds}\mathbb{Q}[S_T \geq K|\mathcal{F}_t].$$

Black-Scholes formula: call option

$$
\begin{aligned}
V_t(G) &= S_t \widetilde{\mathbb{Q}}\left[S_t e^{\int_t^T \left(r(s) + \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widetilde{W}_s} \geq K \middle| \mathcal{F}_t \right] \\
&\quad - K e^{-\int_t^T r(s) ds} \mathbb{Q}\left[S_t e^{\int_t^T \left(r(s) - \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widehat{W}_s} \geq K \middle| \mathcal{F}_t \right] \\
&= S_t \widetilde{\mathbb{Q}}\left[S_t e^{\left(\bar{r} + \frac{\bar{\sigma}^2}{2} \right)(T-t) + \bar{\sigma}(\widetilde{W}_T - \widetilde{W}_t)} \geq K \middle| \mathcal{F}_t \right] \\
&\quad - K e^{-\bar{r}(T-t)} \mathbb{Q}\left[S_t e^{\left(\bar{r} - \frac{\bar{\sigma}^2}{2} \right)(T-t) + \bar{\sigma}(\widehat{W}_T - \widehat{W}_t)} \geq K \middle| \mathcal{F}_t \right] \\
&= S_t N(d_1) - K e^{-\bar{r}(T-t)} N(d_2)
\end{aligned}
$$

with

$$
d_{12} = \frac{\log \frac{S_t}{K e^{-\bar{r}(T-t)}} \pm \frac{\bar{\sigma}^2 (T-t)}{2}}{\bar{\sigma}\sqrt{T-t}}.
$$

Martingale representation and hedging portfolio

- Applying Itô formula to $V_t(G)$, we get:

$$dV_t(G) = N(d_1)dS_t - Ke^{-\bar{r}(T-t)}N(d_2)r(t)dt$$

- Thus, the hedging portfolio contains $N(d_1)$ units of stock and $-Ke^{-\bar{r}(T-t)}N(d_2)$ units of risky asset (borrowing).
- Discounted price satisfies

$$d\widetilde{V}_t(G) = N(d_1)d\widetilde{S}_t = N(d_1)\widetilde{S}_t\sigma(t)d\widehat{W}_t.$$

Example: Black-Scholes price under constraints

What is the Black-Scholes price of a call option when borrowing rate R is higher than lending rate r?

- Since standard BS hedge only requires borrowing, $\overline{V}_0((S_T - K)^+) \leq C^R_{BS}$.
- Assume there exists $x < C^R_{BS}$ and $\pi \in \mathcal{A}$ such that $X^{x,\pi}_T \geq (S_T - K)^+$. Then

$$(S_T - K)^+ \leq X^{x,\pi}_T = x + \int_0^T (X_t - \pi_t)^+ r dt - \int_0^T (X_t - \pi_t)^- R dt + \int_0^T \pi_t \frac{dS_t}{S_t}$$
$$\leq x + \int_0^T (X_t - \pi_t) R dt + \int_0^T \pi_t \frac{dS_t}{S_t}$$

- We found a superhedge at price $x < C^R_{BS}$ in the **standard** Black-Scholes model with rate $R \Rightarrow$ contradiction.
- Thus, $\overline{V}_0((S_T - K)^+) = C^R_{BS}$ and by similar argument, $\underline{V}_0((S_T - K)^+) = C^r_{BS}$